# Chapter 14 Combinatorics - Homework Answers

Chapter revision | Warm-up

**1** **a** $S = \{1, 2, 3, 4, 5, 6\}$ and $E = \{2, 3, 4, 6\}$ **b** $P(E) = \frac{4}{6} = \frac{2}{3}$ **c** The complement is $\{1, 5\}$, so $P(\overline{E}) = 1 - \frac{2}{3} = \frac{1}{3}$.

**2** **a** $\frac{13}{52} = \frac{1}{4}$ **b** $\frac{12}{52} = \frac{3}{13}$ **c** $1 - \frac{3}{13} = \frac{10}{13}$ **d** There are 26 red cards and 4 kings, and the 2 red kings are in both. So $P = \frac{26 + 4 - 2}{52} = \frac{28}{52} = \frac{7}{13}$.

**3** **a** The number who play soccer or tennis is $18 + 14 - 6 = 26$, so $4$ play neither and $P = \frac{4}{30} = \frac{2}{15}$.

**b** Soccer only is $18 - 6 = 12$ and tennis only is $14 - 6 = 8$, so $P = \frac{12 + 8}{30} = \frac{2}{3}$.

**4** **a** $2 \times 2 \times 2 \times 2 = 16$ outcomes **b** $P(\text{first two heads}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ (the last two tosses can be anything) **c**

$P(\text{at least one tail}) = 1 - P(\text{no tails}) = 1 - \frac{1}{16} = \frac{15}{16}$

**5** **a** $\frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$ **b** $P(RB) + P(BR) = \frac{5}{8} \times \frac{3}{7} + \frac{3}{8} \times \frac{5}{7} = \frac{30}{56} = \frac{15}{28}$

**c** $1 - P(\text{both red}) = 1 - \frac{5}{14} = \frac{9}{14}$

# Chapter 14 Combinatorics - Homework Answers

## Lesson 14A | Factorial notation

### Lesson Examples

**C1** **Tutor demo.** **a** $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 6! = 720$ **b** $y' = 4x^3$, $y'' = 4 \times 3x^2$, $y''' = 4 \times 3 \times 2x$, $y^{(4)} = 4 \times 3 \times 2 \times 1 = 4! = 24$ **c** $10! = 10 \times 9! = 3\,628\,800$ and $8! = \dfrac{9!}{9} = 40\,320$

**Your turn.** **a** $7! = 5040$ **b** $y' = 3x^2$, $y'' = 3 \times 2x$, $y''' = 3 \times 2 \times 1 = 3! = 6$ **c** $8! = 8 \times 7! = 40\,320$ and $6! = \dfrac{7!}{7} = 720$

**C2** **Tutor demo.** **a** $9 \times 8 \times 7 = 504$ **b** $(n+3)(n+2)(n+1)$ **c** $n(n-1)(n-2)(n-3)$ **d** $\dfrac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$

**Your turn.** **a** $11 \times 10 \times 9 = 990$ **b** $(n+4)(n+3)(n+2)$ **c** $n(n-1)(n-2)$ **d** $\dfrac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$

**C3** **Tutor demo.** **a** $\dfrac{9 \times 8}{9!} - \dfrac{1}{9!} = \dfrac{71}{9!}$ **b** $\dfrac{n+2}{(n+2)!} + \dfrac{1}{(n+2)!} = \dfrac{n+3}{(n+2)!}$ **c** $10 \times 9! - 9! = 9 \times 9!$ **d** $(n+2)(n+1)! - (n+1)! = (n+1) \times (n+1)!$

**Your turn.** **a** $\dfrac{8 \times 7}{8!} + \dfrac{1}{8!} = \dfrac{57}{8!}$ **b** $\dfrac{n}{n!} - \dfrac{1}{n!} = \dfrac{n-1}{n!}$ **c** $7 \times 6 \times 5! + 5! = 43 \times 5!$ **d** $n(n-1)! + (n-1)! = (n+1) \times (n-1)!$

### Textbook Homework

**1** **a** $8!$ **b** $4!$ **c** $3!$ **d** $101!$ **e** $20!$

**2** **a** 6 **b** 120 **c** 1 **d** 15 **e** 45 **f** 35 **g** 220 **h** 70

**3** **a** 5040 **b** 3 628 800 **c** 1 **d** 15 120 **e** 6720 **f** 2520 **g** 5005 **h** 13 860

**4** **a** $6x^5$ **b** $30x^4$ **c** $120x^3$ **d** $360x^2$ **e** $720x$ **f** $720$ **g** $0$

**5** **a** $n$ **b** $n!$ **c** $1$ **d** $n(n+1)$ **e** $(n+1)(n+2)$ **f** $\dfrac{1}{n(n-1)}$ **g** $\dfrac{n-2}{n}$ **h** $\dfrac{(n-1)!}{n+1}$

**6** **a** $7 \times 7!$ **b** $n \times n!$ **c** $57 \times 6!$ **d** $(n^2+n+1) \times (n-1)!$ **e** $9^2 \times 7!$ **f** $(n+1)^2 \times (n-1)!$

**7** **a** $\dfrac{1+n}{n!}$ **b** $\dfrac{n}{(n+1)!}$ **c** $\dfrac{1-n-n^2}{(n+1)!}$

**8** **a** i $nx^{n-1}$ ii $n(n-1)x^{n-2}$ iii $n!$

iv $n(n-1)(n-2)\cdots(n-k+1)\,x^{n-k} = \dfrac{n!}{(n-k)!}\,x^{n-k}$

**b** i $-1! \times x^{-2}$ ii $2! \times x^{-3}$ iii $-5! \times x^{-6}$ iv $(-1)^n \times n! \times x^{-(n+1)}$

**9** **b** $(n+1)! - 1$

**10** **a** i $2^8$ ii $10^2$ **b** i $2^{97}$ ii $5^{24}$ iii $7^{16}$ iv $13^7$

**12** **a** $\dfrac{1}{2}, \dfrac{1}{3}, \dfrac{1}{8}, \dfrac{1}{30}, \dfrac{1}{144}$ **b** $\dfrac{1}{2}, \dfrac{5}{6}, \dfrac{23}{24}, \dfrac{119}{120}, \dfrac{719}{720}$ **c** $S_n = 1 - \dfrac{1}{(n+1)!}$. The limit is 1.

**d** The sum can be written as

$$\left(\frac{1}{1!} - \frac{1}{2!}\right) + \left(\frac{1}{2!} - \frac{1}{3!}\right) + \left(\frac{1}{3!} - \frac{1}{4!}\right) + \cdots + \left(\frac{1}{n!} - \frac{1}{(n+1)!}\right).$$

**13** **a** $2^{15} \times 15!$ **b** $\dfrac{30!}{2^{15} \times 15!}$ or $\dfrac{29!}{2^{14} \times 14!}$ **c** $\dfrac{2^{30} \times (15!)^2}{30!}$

**14** 0.14%

# Chapter 14 Combinatorics - Homework Answers

## Lesson 14A | Factorial notation

### Past-Paper Homework - Worked Solutions

**PP 1** **Cherrybrook Technology High School, 2020, Year 12 AP4, Mathematics Extension 1, Q1**

Fill the four places of the PIN one at a time. The first digit can be any of the 10 digits. No digit can be repeated, so there are 9 choices left for the second digit, 8 for the third and 7 for the fourth.




$$\text{number of PINs} = 10 \times 9 \times 8 \times 7 = \frac{10!}{6!} = 5040.$$

**Answer (D).** Option (C), $10\,000 = 10^4$, would allow repeated digits. Option (B), 210, is $5040 \div 4!$: it counts the sets of four digits and ignores their order, but 1234 and 4321 are different PINs.

**PP 2** **Ascham School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(e)(i)**

The seven letters of EPSILON are all different. Put P in the first place and S in the last place. The other 5 letters E, I, L, O, N fill the 5 places in between, in $5 \times 4 \times 3 \times 2 \times 1$ ways.




$$\text{number of arrangements} = 5! = 120.$$

**PP 3** **Normanhurst Boys High School, 2022, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(a)(i)**

Unrolling, $(n+2)! = (n+2)(n+1)!$, so $\dfrac{1}{(n+1)!} = \dfrac{n+2}{(n+2)!}$. Hence

$$\begin{aligned}\text{LHS} &= \frac{n+2}{(n+2)!} - \frac{n+1}{(n+2)!} \\ &= \frac{(n+2)-(n+1)}{(n+2)!} \\ &= \frac{1}{(n+2)!} = \text{RHS}.\end{aligned}$$

**PP 4** **Normanhurst Boys High School, 2021, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(e)(i)**

Unrolling, $(n+3)! = (n+3)(n+2)!$, so $\dfrac{1}{(n+2)!} = \dfrac{n+3}{(n+3)!}$. Hence

$$\begin{aligned}\text{LHS} &= \frac{n+3}{(n+3)!} - \frac{n+1}{(n+3)!} \\ &= \frac{(n+3)-(n+1)}{(n+3)!} \\ &= \frac{2}{(n+3)!} = \text{RHS}.\end{aligned}$$

**PP 5** **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q12(c)**

Unrolling, $(n+2)! = (n+2)(n+1)\,n!$, so the equation is $(n+2)(n+1)\,n! = 72\,n!$. Since $n! \neq 0$, divide both sides by $n!$:

$$\begin{aligned}(n+2)(n+1) &= 72 \\ n^2 + 3n + 2 &= 72 \\ n^2 + 3n - 70 &= 0 \\ (n-7)(n+10) &= 0,\end{aligned}$$

so $n = 7$ or $n = -10$. But $n!$ is defined only for whole numbers $n \geq 0$, so $n = -10$ is impossible. Hence $n = 7$.

Check: $9! = 362\,880$ and $72 \times 7! = 72 \times 5040 = 362\,880$.

**PP 6** **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(d)**

Here $n \geq 2$, so that $(n-2)!$ is defined. Write each term in the bracket in terms of $(n-1)!$: unrolling, $n! = n(n-1)!$, so $n!\,n = n^2(n-1)!$. Also $(n-2)!\,(n-1) = (n-1)!$. Hence

$$\begin{aligned}\text{LHS} &= (n+1)\left[n^2(n-1)! + (2n-1)(n-1)! + (n-1)!\right] \\ &= (n+1)(n-1)!\left[n^2 + (2n-1) + 1\right] \\ &= (n+1)(n-1)!\,\left(n^2 + 2n\right) \\ &= (n+1)(n-1)!\,n(n+2) \\ &= (n+2)(n+1)\,n\,(n-1)! \\ &= (n+2)! = \text{RHS}.\end{aligned}$$

# Chapter 14 Combinatorics - Homework Answers

Lesson 14B | Counting ordered selections

## Lesson Examples

**C1** **Tutor demo.** **a** $10 \times 4 \times 7 \times 3 = 840$ **b** $5 \times 21 \times 21 \times 5 = 11\,025$
**Your turn.** **a** $5 \times 8 \times 4 = 160$ **b** $21 \times 5 \times 21 = 2205$

**C2** **Tutor demo.** **a** $4^5 = 1024$ **b** $4^5 - 3^5 = 781$ (the numbers with no 4 use only 1, 2 and 3)
**Your turn.** **a** $6^4 = 1296$ **b** $6^4 - 5^4 = 671$ (the words with no A use only B, C, D, E and F)

**C3** **Tutor demo.** **a** $14 \times 13 \times 12 = {}^{14}\mathrm{P}_3 = 2184$ **b** ${}^{7}\mathrm{P}_6 = \frac{7!}{1!} = 5040$ and ${}^{7}\mathrm{P}_7 = \frac{7!}{0!} = 5040$
**c** $10! = 3\,628\,800$. With 20 people, $20! \doteq 2.4 \times 10^{18}$, about $6.7 \times 10^{11}$ times as many ways, so far more than double. **d** $6 \times 5 \times 4 \times 3 = \frac{6!}{2!} = 360$
**Your turn.** **a** $25 \times 24 = 600$ **b** ${}^{5}\mathrm{P}_4 = 120$ and ${}^{5}\mathrm{P}_5 = 120$ **c** $6! = 720$
**d** $8 \times 7 \times 6 = \frac{8!}{5!} = 336$

**C4** **Tutor demo.** **a** $5 \times 5 \times 8 \times 3 \times 6! = 432\,000$ (Alice, Bob, Carl, Dina, then the last six places) **b** $4 \times 6 \times 5 \times 4 = 480$ (last digit first) **c** $5 \times 5 \times 4 \times 3 = 300$ (first digit first: not 0)
**Your turn.** **a** $3 \times 3 \times 4! = 216$ **b** $3 \times 5 \times 4 = 60$ (last digit first)
**c** $4 \times 4 \times 3 = 48$ (first digit first: not 0)

## Textbook Homework

**1** There are 6: DOG, DGO, ODG, OGD, GOD, GDO

**2** FEG, FGE, FEH, FHE, FEI, FIE, FGH, FHG, FGI, FIG, FHI, FIH

**3** **a** $360$ **b** $720$

**4** **a** $120$ **b** $625$

**5** $60, 36$

**6** $5040$

**7** **a** ${}^{10}\mathrm{P}_3 = 720$ **b** ${}^{5}\mathrm{P}_3 = 60$ **c** ${}^{100}\mathrm{P}_3 = 970\,200$

**8** **a** $9^3 = 729$ **b** $100^3 = 1\,000\,000$ **c** $2^{10} = 1024$

**9** **a** $40\,320$ **b** $336$

**10** **a** $12$ **b** $864$

11 $720$

12 $48$

13 **a** $10^7$ **b** $5 \times 10^6$ **c** $5^7$ **d** $32\,000$

14 **a** $10\,000$ **b** $5040$ **c** $625$ **d** $1000$

15 **a** $3024$ **b** $336$ **c** $1344$ **d** $336$ **e** $1008$

16 **a** $6561$ **b** $729$ **c** $2916$ **d** $729$ **e** $2187$

17 **a** $6\,760\,000$ **b** $3\,276\,000$ **c** $26\,000$ **d** $48$

18 **a** $720$ **b** $120$ **c** $24$ **d** $360$ (half of them)

19 $1728$

20 $24$

21 **a** $120$ **b** $24$ **c** $95$

22 **a** i $64$ ii $32$

23 **a** $96$ **b** $36$ **c** $24$

24 **a** $r = 3$ **b** $n = 3$

25 **a** $30\,000$
**b** $9 \times 9 \times 9 \times 9 \times 3 = 19\,683$ (choose the last digit so that the sum is a multiple of 3)
**c** $9 \times 9 \times 9 \times 9 \times 4 = 26\,244$
**d** $4 \times 9 \times 9 \times 9 \times 3 = 8748$ (choose the last digit first and the first digit last)

# Chapter 14 Combinatorics - Homework Answers

Lesson 14B | Counting ordered selections

## Past-Paper Homework - Worked Solutions

**PP 1** **Normanhurst Boys High School, 2023, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q3**

A number with three digits or fewer is at most 432, so only four-digit numbers can be greater than 2100. Deal with the first digit first.

- First digit 3 or 4: every such number is greater than 2100. There are $2 \times 3 \times 2 \times 1 = 12$ of them.
- First digit 2: the other digits are 1, 3 and 4 in some order, and the smallest of these numbers is 2134. So all $3! = 6$ of them are greater than 2100.
- First digit 1: the number is less than 2000.

$$\text{number of numbers} = 12 + 6 = 18.$$

**Answer (B).** Option (C), $24 = 4!$, counts every four-digit number, including the six that begin with 1.

**PP 2** **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q10(b)(ii)**

The 8 people can stand in a line in $8!$ ways. Swapping Lily and Ben turns each arrangement with Lily on the left of Ben into exactly one arrangement with Ben on the left of Lily, and back again. So exactly half of the arrangements have Lily somewhere to the left of Ben:

$$\text{number of ways} = \frac{8!}{2} = 20\,160.$$

**PP 3** **Gosford High School, 2023, Trial HSC examination, Mathematics Extension 1, Q5**

Selecting 5 of the 6 people is the same as choosing the one person who is left out, so there are 6 selections. For each selection, deal with the restriction first: the oldest of the five goes at the left or the right end (2 ways), and the second oldest takes the other end (1 way). The other three people fill the middle three places in $3!$ ways.

$$\text{number of ways} = 6 \times 2 \times 1 \times 3! = 72.$$

**Answer C.** Option A, $720 = {}^6\mathrm{P}_5$, ignores the restriction on the ends.

**PP 4** **Hurlstone Agricultural High School, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q11(b)**

The eight letters C, O, N, S, I, D, E, R are all different, so there are $|S| = 8!$ equally likely arrangements. In the arrangements that begin with SI, the first two places are fixed and the other 6 letters fill the last 6 places, so $|E| = 6!$. Hence

$$P(\text{begins with SI}) = \frac{6!}{8!} = \frac{1}{8 \times 7} = \frac{1}{56}.$$

**PP 5** **Girraween High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q14(c)**

The numbers may have from one to five digits. A number with three digits or fewer is less than 1000, so it is not counted.

**Four-digit numbers.** Deal with the first digit first: it must be 7 or 8 (2 ways). The other three places are then filled from the four digits left:

$$\text{four-digit numbers} = 2 \times 4 \times 3 \times 2 = 48.$$

**Five-digit numbers.** Every five-digit number is greater than 6000, so all of them are counted:

$$\text{five-digit numbers} = 5! = 120.$$

Hence the number of numbers greater than 6000 is $48 + 120 = 168$.

**PP 6** **Ascham School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(e)(iii)**

The seven letters of EPSILON are all different, so there are $7!$ arrangements. Swapping N and L pairs each arrangement with N to the left of L with exactly one arrangement with L to the left of N. So half of the arrangements are counted:

$$\text{number of arrangements} = \frac{7!}{2} = 2520.$$

**PP 7** **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q12(d)**

Let each of the three equal ratios be $k$. Then $a = \dfrac{{}^n\mathrm{P}_{r-1}}{k}$, $b = \dfrac{{}^n\mathrm{P}_r}{k}$ and $c = \dfrac{{}^n\mathrm{P}_{r+1}}{k}$.

Using ${}^n\mathrm{P}_r = \dfrac{n!}{(n-r)!}$ and unrolling,

$$\frac{b}{a} = \frac{{}^n\mathrm{P}_r}{{}^n\mathrm{P}_{r-1}} = \frac{n!}{(n-r)!} \times \frac{(n-r+1)!}{n!} = \frac{(n-r+1)!}{(n-r)!} = n-r+1,$$

$$\frac{c}{b} = \frac{{}^n\mathrm{P}_{r+1}}{{}^n\mathrm{P}_r} = \frac{n!}{(n-r-1)!} \times \frac{(n-r)!}{n!} = \frac{(n-r)!}{(n-r-1)!} = n-r.$$

Hence $\dfrac{b}{a} = \dfrac{c}{b} + 1$, that is,

$$\frac{b}{a} = \frac{b+c}{b}$$

$$b^2 = a(b+c).$$

**PP 8** **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(c)(i)(ii)**

The first digit is 1. Exactly one pair of identical digits means that one digit appears exactly twice and the other two digits are different from it and from each other. The digit 0 is allowed in the last three places.

(i) The repeated digit is 1, so there is one more 1. Deal with it first: it goes in one of the last 3 places (3 ways). The two remaining places hold two different digits other than 1, chosen in order from the 9 digits 0, 2, 3, ..., 9:

$$\text{number of numbers} = 3 \times 9 \times 8 = 216.$$

(ii) If the repeated digit is not 1, then the 1 appears only at the front. The last three places hold one digit once and another digit twice, and neither of them is 1. Place the digit that appears once first: its position (3 ways) and its value (9 ways, any digit except 1). The repeated digit fills the other two places, and it is not 1 and not the digit just placed (8 ways). So there are $3 \times 9 \times 8 = 216$ numbers in this case. Adding the case from part (i),

$$\text{total} = 216 + 216 = 432.$$

# Chapter 14 Combinatorics - Homework Answers

## Lesson 14C | Ordered selections and grouping

### Lesson Examples

**C1** **Tutor demo. a** $6! \times 2! \times 2! = 2880$ (six groups: the couple, the two girls and four single boys) **b** $6! \times 2! = 1440$ (glue A and I)
**Your turn. a** $4! \times 2! \times 2! = 96$ **b** $4! \times 2! = 48$

**C2** **Tutor demo. a** $6! - 2 \times 5! = 480$ **b** $9! - 2 \times 8! = 282\,240$
**Your turn. a** $5! - 2 \times 4! = 72$ **b** $6! - 2 \times 5! = 480$

**C3** **Tutor demo. a** $200 + 60 = 260$ (no overlap; zero counts as a multiple of 5)
**b** $350 + 120 - 30 = 440$ (the overlap is 305, 315, ..., 595)
**c** $24 + 12 = 36$ (four-digit numbers $4!$, plus three-digit numbers beginning with 5 or 7, $2 \times 3 \times 2$)
**Your turn. a** $150 + 20 = 170$ **b** $300 + 140 - 30 = 410$ **c** $6 + 2 = 8$ (all $3!$ three-digit numbers, plus 73 and 75)

### Textbook Homework

**1** **a** $5! \times 2! = 240$ **b** $2! \times 2! = 4$ **c** $3! \times 2! = 12$ **d** $5! \times 3! = 720$

**2** **a** $4! \times 2! = 48$ **b** $4! = 24$

**3** $3! \times 2! \times 2! = 24$

**4** $3! \times 3! \times 6! \times 4! = 622\,080$

**5** **a** $2 \times 3 \times 3 \times 2 \times 1 + 3 \times 4! = 108$ $(36 + 72)$ **b** $5! - 2 \times 3! = 108$ $(120 - 12)$

**6** If the father drives, there are $2 \times 2 \times 1$ ways to arrange the seating. If the mother drives, there are $1 \times 2 \times 1$ ways. Thus there are $6$ ways in total.

**7** Three-digit numbers: $3 \times 2 \times 1 = 6$. Two-digit numbers: $2 \times 2 = 4$. The total is $10$.

**8** **a** $144$ **b** $120$ **c** $144$ **d** $2520$ (half of the total)

**9** **a** $720$ **b** $720$ **c** $4320$

**10** **a** $24$ **b** $240$

**11** $2 + 4 + 8 + \cdots + 1024 = 2046$

**12** **a** $1152$ **b** $1152$

**13** **a** $720$ **b** $120$ **c** $1680$ **d** $4200$ **e** $960$ **f** $480$

**14** **a** $5040$ **b** $4320$ **c** $720$ **d** $144$ **e** $720$ **f** $960$ **g** $1440$

**15** **a** $7^7$ **b** $6 \times 7^6$ **c** $7^6$ **d** $3^4 \times 4^3 + 4^4 \times 3^3 = 7 \times 12^3$

**16** **a** i $3\,628\,800$ ii $725\,760$ iii $725\,760$ iv $2\,257\,920$
**b** i $2(n-1)!$ ii $2(n-1)!$ iii $(n-2)(n-3)(n-2)!$

**17** $8640$

**18** **a** $40\,320$ **b** $20\,160$ **c** $17\,280$

**19** **a** $5040$ **b** $20\,160$

**20** **a** $5^5$ ways **b** $5! = 120$ ways **c** $5 \times 4^3 = 320$ ways

**21** **a** $133$ **b** $104$ **c** $29$ **d** $56$

**22** **a** $D(1) = 0$, $D(2) = 1$, $D(3) = 2$, $D(4) = 9$
**c** $D(5) = 44$, $D(6) = 265$, $D(7) = 1854$, $D(8) = 14\,833$

# Chapter 14 Combinatorics - Homework Answers

Lesson 14C | Ordered selections and grouping

## Past-Paper Homework - Worked Solutions

**PP 1** **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q1**

The three girls form one group, and each of the 5 boys is a group on his own, so there are 6 groups. Order the 6 groups, then order the girls within their group (Box 8):

$$\text{number of ways} = 6! \times 3! = 4320.$$

**Answer C.** Option A, $40\,320 = 8!$, ignores the restriction. Option B, $720 = 6!$, forgets to order the girls within their group.

**PP 2** **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q3**

The seven letters O, L, Y, M, P, I, C are all different. Glue the C and the L together as one group. Then there are 6 groups (CL, O, Y, M, P, I), which can be ordered in $6!$ ways, and the C and L can be ordered within their group in $2!$ ways:

$$\text{number of arrangements} = 6! \times 2! = 1440.$$

**Answer (D).** Option (B), $720 = 6!$, forgets that the pair can be CL or LC.

**PP 3** **Ascham School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(e)(ii)**

The seven letters of EPSILON are all different, and the vowels are E, I and O. Glue the three vowels together as one group, order the 5 groups (the vowels, P, S, L, N) in $5!$ ways, then order the vowels within their group in $3!$ ways:

$$\text{number of arrangements} = 5! \times 3! = 720.$$

**PP 4** **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q9**

Deal with the boys first: seat them in a row in $3!$ ways. This leaves 4 gaps where a girl can sit, one at each end and one between each pair of boys (each ☐ below is a gap):

☐ B ☐ B ☐ B ☐

The girls sit apart exactly when no two girls are in the same gap. The first girl chooses a gap in 4 ways, the second in 3 ways and the third in 2 ways:

$$\text{number of ways} = 3! \times 4 \times 3 \times 2 = 144.$$

**Answer (C).** Option (A), $72 = 2 \times 3! \times 3!$, counts only the seatings in which the boys and girls alternate (GBGBGB and BGBGBG). It misses seatings such as GBBGBG.

**PP 5** **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q10**

The dice are different colours, so an outcome is an ordered triple of scores, and there are $|S| = 6^3 = 216$ equally likely outcomes. The only perfect squares among the scores are 1 and 4. Use cases based on how many of the three scores are equal. The cases do not overlap.

- **All three scores equal**, $a, a, a$: the product $a^3$ is a square only for $a = 1$ and $a = 4$ ($4^3 = 64 = 8^2$). This gives 2 outcomes.
- **Exactly two scores equal**, $a, a, b$ with $b \neq a$: the product $a^2 b$ is a square exactly when $b$ is a square, so $b = 1$ or $b = 4$. For each $b$ there are 5 choices of $a$, and the odd score $b$ can be on any of the 3 dice. This gives $2 \times 5 \times 3 = 30$ outcomes.
- **All three scores different**: the only such set of scores is $\{2, 3, 6\}$, with product $36 = 6^2$. (A single 5 makes the product a multiple of 5 but not of 25, so the scores come from 1, 2, 3, 4 and 6. Since 1 and 4 are squares, the product of the scores taken from 2, 3 and 6 must be a square. Three different scores cannot all come from 1 and 4, and of the products $2,\ 3,\ 6,\ 2 \times 3,\ 2 \times 6,\ 3 \times 6,\ 2 \times 3 \times 6$ only $2 \times 3 \times 6 = 36$ is a square.) The three scores can be placed on the dice in $3! = 6$ ways.

$$P(\text{perfect square}) = \frac{2 + 30 + 6}{216} = \frac{38}{216}.$$

**Answer (D).**

**PP 6** **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q11(d)**

Without restriction the seven people can board in $7!$ orders. Count the unacceptable orders, in which Melania gets on immediately following Donald. Glue Donald and Melania together, in the order DM only, so there are 6 groups to order, in $6!$ ways. Subtract:

$$\begin{aligned} \text{number of ways} &= 7! - 6! \\ &= 4320. \end{aligned}$$

Note that the pair is not multiplied by $2!$ here: "Melania immediately following Donald" allows only the order DM.

**PP 7** **Pymble Ladies' College, 2024, HSC Trial Examination Term 3, Mathematics Extension 1, Q11(b)**

Without restriction the 8 exams can be scheduled in $8!$ ways. Count the complementary situation, where Extension 1 and Extension 2 are consecutive: glue them together, order the 7 groups in $7!$ ways, and order the two exams within their group in $2!$ ways. Subtract:

$$\begin{aligned} \text{number of ways} &= 8! - 2 \times 7! \\ &= 30\,240. \end{aligned}$$

**PP 8** **Blacktown Boys' High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(c)(i)(ii)**

Think of the teacher choosing the four batteries one at a time: the 1st and 2nd go in the projector remote, and the 3rd and 4th go in the air-conditioner remote. The batteries are all different, so the number of equally likely ordered selections is

$$|S| = 11 \times 10 \times 9 \times 8 = 7920.$$

This ordered model gives the same probabilities as unordered selection: each choice of an unordered pair for each remote matches exactly $2! \times 2! = 4$ ordered selections, so each count of selections of all four batteries (including the total) is 4 times the unordered count, and the 4 cancels.

(i) Both remotes work when all four batteries are good. There are $7 \times 6 \times 5 \times 4 = 840$ such selections, so

$$P(\text{both work}) = \frac{840}{7920} = \frac{7}{66}.$$

(ii) The projector remote works when its two batteries are good: $7 \times 6 = 42$ ways. Then 9 batteries are left, of which 5 are good. The air-conditioner remote does not work, so count the complementary situation: its two batteries can be chosen in $9 \times 8 = 72$ ways, of which $5 \times 4 = 20$ are both good. So there are $72 - 20 = 52$ ways for it to fail. Hence

$$\begin{aligned} P(\text{only the projector works}) &= \frac{42 \times 52}{7920} \\ &= \frac{2184}{7920} \\ &= \frac{91}{330}. \end{aligned}$$

# Chapter 14 Combinatorics - Homework Answers

## Lesson 14D | Ordered selections with identical elements

### Lesson Examples

**C1**

**Tutor demo.**

**a** $\dfrac{9!}{4! \times 5!} = 126$

**b** $\dfrac{9!}{3! \times 1! \times 3! \times 2!} = 5040$ (3 plain mugs, the chipped mug, 3 old cups, 2 new cups)

**c** $\dfrac{6!}{3! \times 2!} = 60$ (three Ps and two Es)

**Your turn.**

**a** $\dfrac{6!}{2! \times 4!} = 15$

**b** $\dfrac{6!}{1! \times 1! \times 2! \times 2!} = 180$ (plain vase, chipped vase, 2 old jars, 2 new jars)

**c** $\dfrac{5!}{2! \times 2!} = 30$ (two Ts and two Os)

**C2**

**Tutor demo.**

**a** $\dfrac{8!}{5! \times 3!} = 56$

**b** $2^8 = 256$

**c** $\dfrac{6!}{4! \times 2!} = 15$ (each path is a word with four Rs and two Ds)

**Your turn.**

**a** $\dfrac{6!}{4! \times 2!} = 15$

**b** $2^6 = 64$

**c** $\dfrac{6!}{3! \times 3!} = 20$ (three Rs and three Ds)

**C3** **Tutor demo.** Omit each letter in turn.

If D or S is omitted (2 cases), one pair remains: $\frac{6!}{2!} \times 2 = 720$.

If A, R or E is omitted (3 cases), two pairs remain: $\frac{6!}{2! \times 2!} \times 3 = 540$.

Total $720 + 540 = 1260$.

**Your turn.** Omit each letter in turn.

If E or T is omitted (2 cases), one pair remains: $\frac{5!}{2!} \times 2 = 120$.

If L or R is omitted (2 cases), two pairs remain: $\frac{5!}{2! \times 2!} \times 2 = 60$.

Total $120 + 60 = 180$.

Textbook Homework

**1** **a** 3 **b** 12 **c** 120 **d** 6720 **e** 10 080 **f** 90 720 **g** 4 989 600 **h** 45 360 **i** 25 740

**2** $\frac{6!}{3! \times 2!} = 60$

**3** **a** 6 **b** 15 **c** 20

**4** **a** 40 320 **b** 8 **c** 56 **d** 560

**5** **a** 56 **b** 20

**6** **a** 56 **b** 5

**7** **a** 60 **b** 24 **c** 36 **d** 30 (half of them)

**8** **a** i 180 ii 60 iii 120 iv 24 **b** 40

**9** **a** 90 720 **b** 720 **c** 720 **d** 45 360 (half of them)

**10** 2 721 600

**11** **a** 1024 **b** 256 **c** 45 **d** 252 **e** 56 **f** 512 **g** 8 **h** 70

**12** **a** 60 **b** 60

**13** **a** 120 **b** 60

**14** **a** 453 600 **b** 90 720 **c** 5040 **d** 10 080 **e** 80 640 **f** 282 240 **g** 15 120

**15** **a** 3 628 800 **b** 4

**16** **a** $2520$ **b** $720$ **c** i $600$ ii $480$ iii $360$ iv $240$ v $120$

**d** $840$. Insert the letters U, M, T and R successively into the word EGE. Alternatively, the answer is one third of all arrangements.

**e** $210$ **f** $420$

**17** $1\,995\,840$

**18** $864$

**19** **a** $4!$

**b** Each is $3! = 6$, so subtract $4 \times 3! = 24$.

**c** $2!$ permutations leave A and B unmoved, and there are 6 pairs of letters, so add $6 \times 2! = 12$.

**d** $D(4) = 1 \times 4! - 4 \times 3! + 6 \times 2! - 4 \times 1! + 1 \times 0! = 9$

**g** The ratio of the number of permutations of $n$ distinct letters to the number of derangements of them converges to $e$ as $n \to \infty$. Thus, for example, if a long queue is formed at random and then rearranged into alphabetical order, the probability that no-one remains in his or her original position is $\dfrac{1}{e}$.

# Chapter 14 Combinatorics - Homework Answers

Lesson 14D | Ordered selections with identical elements

## Past-Paper Homework - Worked Solutions

**PP 1** **Knox Grammar School, 2022, Higher School Certificate Trial Examination, Mathematics Extension 1, Q7**

EXCELLENT has 9 letters: three Es, two Ls, and X, C, N, T once each. By Box 10,

$$\text{number of arrangements} = \frac{9!}{3! \times 2!} = 30\,240.$$

**Answer B.** Option D, $362\,880 = 9!$, treats all nine letters as different. Option C, $60\,480 = \dfrac{9!}{3!}$, allows for the repeated Es but not for the repeated Ls.

**PP 2** **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q11(a)**

Only the colours matter, so the row of chairs is a six-letter word using the colours black, black, blue, green, yellow and grey. The two black chairs cannot be told apart, so by Box 10

$$\text{number of arrangements} = \frac{6!}{2!} = 360.$$

Note that the two black chairs are not glued together: other chairs may stand between them.

**PP 3** **Sydney Technical High School, 2023, Extension 1 Mathematics, Trial HSC, Q1**

SPUDDLE has 7 letters, with two Ds. Deal with the restriction first: the P goes in the first place in 1 way. The remaining six letters S, U, D, D, L, E fill the other six places, with two Ds alike, so

$$\text{number of arrangements} = 1 \times \frac{6!}{2!} = 360.$$

**Answer B.** Option A, $\dfrac{7!}{2!}$, is the number of arrangements with no restriction.

**PP 4** **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q12(a)**

TABLECLOTH has 10 letters, with two Ts and two Ls, and the other six letters A, B, E, C, O, H once each. By Box 10,

$$\text{number of arrangements} = \frac{10!}{2! \times 2!} = 907\,200.$$

**PP 5** **Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q11(c)**

The four letters of MILO are all different, so MILO has $4! = 24$ arrangements.

SAVVAS has 6 letters: two Ss, two As and two Vs. By Box 10 it has

$$\frac{6!}{2! \times 2! \times 2!} = 90 \text{ arrangements.}$$

Since $90 > 24$, SAVVAS has more arrangements.

**PP 6** **Baulkham Hills High School, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q8**

EXTENSION has 9 letters: two Es, two Ns, and X, T, S, I, O once each. Glue the two Es together as one group EE. Because the two Es are identical, the group can be ordered in only 1 way. There are now 8 items to arrange: EE, X, T, N, S, I, O, N, with two Ns alike. By Box 10,

$$\text{number of arrangements} = \frac{8!}{2!} = 20\,160.$$

**Answer D.** Option C, $\dfrac{8!}{2!\,2!}$, wrongly divides by $2!$ for the Es as well. Once the Es are glued together they form a single item, so there is nothing left to overcount.

**PP 7** **St George Girls High School, 2020, Trial HSC Examination, Mathematics Extension 1, Q11(c)(i)(ii)**

STATISTICS has 10 letters: three Ss, three Ts, two Is, and A and C once each.

(i) By Box 10,

$$\text{number of arrangements} = \frac{10!}{3! \times 3! \times 2!} = 50\,400.$$

(ii) Glue the A and C together as one group. The group can be ordered in $2!$ ways (AC or CA). There are now 9 items: the group, three Ss, three Ts and two Is. Hence

$$\begin{aligned}\text{number of arrangements} &= 2! \times \frac{9!}{3! \times 3! \times 2!} \\ &= 10\,080.\end{aligned}$$

**PP 8** **Normanhurst Boys High School, 2021, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q3**

CALCULUS has 8 letters: two Cs, two Ls, two Us, and A and S once each. The number of different arrangements is

$$|S| = \frac{8!}{2! \times 2! \times 2!} = 5040.$$

These arrangements are equally likely: if the 8 letters are shuffled at random, each different arrangement comes from exactly $2! \times 2! \times 2! = 8$ of the $8!$ orders of the letters.

For a C at the start and at the end, both Cs are used up. The six letters A, L, U, L, U, S fill the middle six places, with two Ls and two Us alike:

$$|E| = \frac{6!}{2! \times 2!} = 180.$$

$$P(\text{C at the start and the end}) = \frac{180}{5040} = \frac{1}{28}.$$

**Answer (C).**

**PP 9** **Ascham School, 2020, Trial HSC Examination, Mathematics Extension 1, Q2**

The six cards A, A, C, H, M, S include two As. The number of different words the cards can form is

$$\frac{6!}{2!} = 360.$$

These words are equally likely, since each one comes from exactly $2! = 2$ of the $6!$ orders of the cards (the two A cards can be swapped). Only one of the words is ASCHAM, so

$$P(\text{ASCHAM}) = \frac{1}{360}.$$

**Answer B.** Option C, $\frac{1}{720}$, forgets that the two A cards can be swapped.

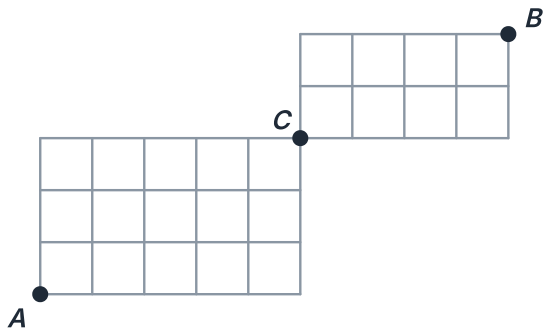
**PP 10** **Normanhurst Boys High School, 2022, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(e)**

The grid is 5 unit squares wide and 3 unit squares high. Every path from $A$ to $B$ is made of 5 steps right (R) and 3 steps down (D), in some order. So each path is an eight-letter word with five Rs and three Ds, and each such word gives exactly one path. By Box 11,

$$\text{number of paths} = \frac{8!}{5! \times 3!} = 56.$$

**PP 11** **St Aloysius' College, 2024, Trial Higher School Certificate Examination, Mathematics Extension 1, Q8**

The two grids meet only at the single point $C$, so every path from $A$ to $B$ passes through $C$.




From $A$ to $C$ a path has 5 steps right and 3 steps up: $\dfrac{8!}{5! \times 3!} = 56$ paths (Box 11). From $C$ to $B$ a path has 4 steps right and 2 steps up: $\dfrac{6!}{4! \times 2!} = 15$ paths. Any first part can be followed by any second part, so

$$\textbf{number of paths} = 56 \times 15 = 840.$$

**Answer (B).**

**PP 12** **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q10**

Line up the 10 players in a fixed order, and give each player a label for what happens to them: 1st, 2nd, 3rd or 4th; F for a finalist who is not placed; or O for a player who is not selected. Each way of carrying out the process gives exactly one list of labels, and each list using the labels

$$\textbf{1st, 2nd, 3rd, 4th, F, F, O, O, O, O}$$

gives exactly one way. This is a ten-letter word with two Fs alike and four Os alike, so by Box 10

$$\textbf{number of ways} = \frac{10!}{4! \times 2!} = 75\,600.$$

**Answer (C).** Option (D), $\dfrac{10!}{6!}$, only places 4 of the 10 players and forgets to choose the other 2 finalists.

# Chapter 14 Combinatorics - Homework Answers

## Lesson 14E | Counting unordered selections

### Lesson Examples

**C1**

**Tutor demo.**

**a** $2^6 = 64$

**b** ${}^6\mathrm{C}_4 = \dfrac{6!}{4! \times 2!} = 15$

**c** ${}^9\mathrm{C}_4 = \dfrac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126$

**d** ${}^n\mathrm{C}_4 = \dfrac{n(n-1)(n-2)(n-3)}{24}$

**e** ${}^{15}\mathrm{C}_4 = \dfrac{15 \times 14 \times 13 \times 12}{1 \times 2 \times 3 \times 4} = 1365 = 3 \times 5 \times 7 \times 13$

**Your turn.**

**a** $2^4 = 16$

**b** {A, B}, {A, C}, {A, D}, {B, C}, {B, D}, {C, D}, and ${}^4\mathrm{C}_2 = \dfrac{4!}{2! \times 2!} = 6$

**c** ${}^7\mathrm{C}_3 = \dfrac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$

**d** ${}^n\mathrm{C}_2 = \dfrac{n(n-1)}{2}$

**e** ${}^{12}\mathrm{C}_5 = \dfrac{12 \times 11 \times 10 \times 9 \times 8}{1 \times 2 \times 3 \times 4 \times 5} = 792 = 2^3 \times 3^2 \times 11$

**C2**

**Tutor demo.**

**a** ${}^9\mathrm{C}_2 = \dfrac{9!}{2! \times 7!}$ and ${}^9\mathrm{C}_7 = \dfrac{9!}{7! \times 2!}$, which are equal.

**b** ${}^{20}\mathrm{C}_{18} = {}^{20}\mathrm{C}_2 = \dfrac{20 \times 19}{2} = 190$

**c** ${}^9\mathrm{C}_7 = {}^9\mathrm{C}_2 = 36$ (choose the two people who will not make the tea)

**Your turn.**

**a** ${}^8\mathrm{C}_3 = \dfrac{8!}{3! \times 5!}$ and ${}^8\mathrm{C}_5 = \dfrac{8!}{5! \times 3!}$, which are equal.

**b** ${}^{15}\mathrm{C}_{13} = {}^{15}\mathrm{C}_2 = \dfrac{15 \times 14}{2} = 105$

**c** ${}^7\mathrm{C}_5 = {}^7\mathrm{C}_2 = 21$ (choose the two people who will not make the tea)

## C3

**Tutor demo.**

**a** ${}^{12}\mathrm{C}_4 = 495$

**b** ${}^{10}\mathrm{C}_3 = 120$ (three more from the ten people other than Maria and Alex)

**c** ${}^{5}\mathrm{C}_2 \times {}^{7}\mathrm{C}_2 = 10 \times 21 = 210$

**d** ${}^{7}\mathrm{C}_1 \times {}^{5}\mathrm{C}_3 + {}^{5}\mathrm{C}_4 = 70 + 5 = 75$ (three women and one man, or four women)

**Your turn.**

**a** ${}^{9}\mathrm{C}_4 = 126$

**b** ${}^{7}\mathrm{C}_3 = 35$ (three more from the seven people other than Maria and Alex)

**c** ${}^{4}\mathrm{C}_2 \times {}^{5}\mathrm{C}_2 = 6 \times 10 = 60$

**d** ${}^{5}\mathrm{C}_1 \times {}^{4}\mathrm{C}_3 + {}^{4}\mathrm{C}_4 = 20 + 1 = 21$ (three women and one man, or four women)

## C4

**Tutor demo.**

**a** $2^7 - \left({}^{7}\mathrm{C}_1 + {}^{7}\mathrm{C}_0\right) = 128 - 8 = 120$

**b** Use the 6-member set {3, 6, 12, 15, 18, 21}: $2^6 - \left({}^{6}\mathrm{C}_1 + {}^{6}\mathrm{C}_0\right) = 64 - 7 = 57$

**c** 21 is already chosen. Choose at least one more from {3, 6, 12, 15, 18}: $2^5 - 1 = 31$

**Your turn.**

**a** $2^5 - \left({}^{5}\mathrm{C}_1 + {}^{5}\mathrm{C}_0\right) = 32 - 6 = 26$

**b** Use the 4-member set {5, 15, 20, 25}: $2^4 - \left({}^{4}\mathrm{C}_1 + {}^{4}\mathrm{C}_0\right) = 16 - 5 = 11$

**c** 25 is already chosen. Choose at least one more from {5, 15, 20}: $2^3 - 1 = 7$

## C5

**Tutor demo.**

**a** ${}^{8}\mathrm{C}_3 = 56$

**b** ${}^{8}\mathrm{C}_6 \times {}^{5}\mathrm{C}_2 = 28 \times 10 = 280$

**c** i ${}^{8}\mathrm{C}_6 \times 3 = 84$ ii $280 - 84 = 196$

**Your turn.**

**a** ${}^{7}\mathrm{C}_3 = 35$

**b** ${}^{7}\mathrm{C}_6 \times {}^{5}\mathrm{C}_2 = 7 \times 10 = 70$

**c** i ${}^{7}\mathrm{C}_6 \times 3 = 21$ ii $70 - 21 = 49$

## Textbook Homework

**1** There are $^5\mathrm{C}_2 = 10$ possible combinations: PQ, PR, PS, PT, QR, QS, QT, RS, RT and ST.

**2** **a** $21$ **b** $35$ **c** $15$ **d** $126$

**3** **a** i $45$ ii $45$
**b** $^{10}\mathrm{C}_2 = {}^{10}\mathrm{C}_8$, and in general $^n\mathrm{C}_r = {}^n\mathrm{C}_{n-r}$. Choosing the two people who are chosen is the same as choosing the eight people who are not.

**4** **a** $44\,352$ **b** $34\,650$

**5** **a** $70$ **b** $36$ **c** $16$ **d** $1$ **e** $69$

**7** **a** $2002$ **b** $56$ **c** $6$ **d** $840$ **e** $420$ **f** $1316$ **g** $715$ **h** $1287$

**8** **a** $70$ **b** $5$ **c** $35$

**9** **a** $792$ **b** $462$ **c** $120$ **d** $210$ **e** $420$

**10** **a** i $252$ ii $126$. The number cannot begin with a zero.
**b** In each part, once the five digits have been selected, they can only be arranged in one way.

**11** $13\,860$

**12** **a** $1\,745\,944\,200$ **b** $413\,513\,100$

**13** **a** $45$ **b** $120$ **c** $36$ **d** $8$

**14** **a** $10$ **b** $110$

**15** **a** $65\,780$ **b** $1287$ **c** $48$ **d** $22\,308$ **e** $288$ **f** $3744$

**16** **a** i $^6\mathrm{C}_1 + {}^6\mathrm{C}_2 = 21$ ii $^5\mathrm{C}_2 = 10$ (choose the two people to go in the same group as one particular person)
**b** i $4$ ii $3$ **c** i $92$ ii $35$

**17** **a** $2$ **b** $5$ **c** $35$ **d** $^n\mathrm{C}_2 - n$

**18** **a** $220$ **b** $9240$ **c** i $2772$ ii $6468$

**19** **a** $1024$ **b** $968$ **c** $466$ **d** $247$

**20** **a** $16$ **b** $20$ **c** $12$ **d** $8$ **e** $5$

**21** **a** $252$ **b** $126$

**22** **a** $315$ **b** $210$

23 **a** $12$ **b** $49$ **c** $120$ **d** $(a+1)(b+1)(c+1)$

24 **a** $30$ **b** $24$

25 **a** i $210$ ii $90$ iii $126$ iv $126$

26 $5151$

27 $1360$

28 **a** $3$ **b** $315$ **c** i $155\,925$ ii $10\,800$

29 **b** For $n = 1, 2, 3, 4, 5, 6$ the numbers of allowed arrangements are $0$ (ratio undefined), $2$ (ratio $3$), $30$ (ratio $3$), $864$ (ratio $2.917$), $39\,480$ (ratio $2.872$), $2\,631\,600$ (ratio $2.844$).

30 **a** i ${}^{a}\mathrm{C}_{0}$ ii ${}^{a+1}\mathrm{C}_{1}$ iii ${}^{a+2}\mathrm{C}_{2}$ iv ${}^{a+3}\mathrm{C}_{3}$ v ${}^{a+r}\mathrm{C}_{r}$
**b** Add them up, for $r = 0, 1, \ldots, b$.
**c** There are $a+1$ 0s and $b$ 1s, and the total length is $a+b+1$.
**d** Go to the last 0 in the string, and remove it and any 1s that follow it. What remains is a string with $a$ 0s and at most $b$ 1s. When the process in part **c** is applied to the shortened string, the original string returns.
**e** The one-to-one correspondence in parts **c** and **d** shows that $N$ is the number of strings with $a+1$ 0s and $b$ 1s, which is ${}^{a+b+1}\mathrm{C}_{b}$.
**f** This follows from parts **b** and **e**.

# Chapter 14 Combinatorics - Homework Answers

Lesson 14E | Counting unordered selections

## Past-Paper Homework - Worked Solutions

**PP 1** **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q6**

A group of 4 people is an unordered selection, so

$$\text{number of ways} = {}^{20}\mathrm{C}_4 = \frac{20 \times 19 \times 18 \times 17}{4 \times 3 \times 2 \times 1} = 4845.$$

**Answer A.** Option C, $116\,280 = {}^{20}\mathrm{P}_4$, counts ordered selections, so it counts each group $4! = 24$ times.

**PP 2** **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q6**

'At least one of each sex' is a warning, so consider the complement. With no restriction there are ${}^{10}\mathrm{C}_3 = 120$ committees. The committees that break the condition are all male, ${}^{5}\mathrm{C}_3 = 10$, or all female, ${}^{5}\mathrm{C}_3 = 10$. Hence

$$\text{number of committees} = 120 - 10 - 10 = 100.$$

Check by cases: two men and one woman give ${}^{5}\mathrm{C}_2 \times {}^{5}\mathrm{C}_1 = 50$, and one man and two women give another 50.

**Answer (B).**

**PP 3** **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q3**

The team has 5 members with at least two from each year, so it has either 2 from Year 11 and 3 from Year 12, or 3 from Year 11 and 2 from Year 12.

$$\begin{aligned}\text{number of teams} &= {}^{10}\mathrm{C}_2 \times {}^{15}\mathrm{C}_3 + {}^{10}\mathrm{C}_3 \times {}^{15}\mathrm{C}_2 \\ &= 45 \times 455 + 120 \times 105 \\ &= 33\,075.\end{aligned}$$

**Answer C.**

**PP 4** **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q6**

The captain and the vice captain have different roles, so choosing them is an ordered selection: the captain in $n$ ways, then the vice captain in $n-1$ ways. The other $r-2$ members of the team have no special roles, so they are an unordered selection from the $n-2$ students who are left: $\binom{n-2}{r-2}$ ways. Hence

$$\text{number of ways} = n(n-1)\binom{n-2}{r-2}.$$

**Answer D.** Another correct method chooses the team first, in $\binom{n}{r}$ ways, and then its captain and vice captain in $r(r-1)$ ways. This gives $r(r-1)\binom{n}{r}$. That form is not printed as an option, but it is equal to option D. Option C uses $\binom{n-2}{r}$, which chooses too many other members.

**PP 5** **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q12(d)(i)(ii)**

(i) Choose the two boys from 14 and, separately, the two girls from 13:

$$\text{number of ways} = {}^{14}\mathrm{C}_2 \times {}^{13}\mathrm{C}_2 = 91 \times 78 = 7098.$$

(ii) Henry is one of the boys and Olivia is one of the girls. One more boy is chosen from the other 13 boys, and one more girl from the other 12 girls:

$$\text{number of ways} = {}^{13}\mathrm{C}_1 \times {}^{12}\mathrm{C}_1 = 13 \times 12 = 156.$$

**PP 6** **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q11(e)**

With no restriction there are ${}^{10}\mathrm{C}_5 = 252$ committees. The committees that break the condition contain both Peter and May. For these, the other 3 members are chosen from the other 8 candidates in ${}^{8}\mathrm{C}_3 = 56$ ways. Hence

$$\text{number of committees} = {}^{10}\mathrm{C}_5 - {}^{8}\mathrm{C}_3 = 252 - 56 = 196.$$

**PP 7** **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q13(a)**

Any 3 of the 9 points make a triangle, unless the 3 points are collinear. The only collinear sets of three points are sets of three of the four points on the line. Hence

$$\text{number of triangles} = {}^{9}\mathrm{C}_3 - {}^{4}\mathrm{C}_3 = 84 - 4 = 80.$$

**PP 8** **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(e)(i)(ii)**

(i) The team is any 12 of the $8+16=24$ students:

$$\text{number of ways} = {}^{24}\mathrm{C}_{12} = 2\,704\,156.$$

(ii) In a team of 12, more left-handed than right-handed means at least 7 left-handed students. There are only 8, so the team has 7 left and 5 right, or 8 left and 4 right:

$$\begin{aligned}\text{number of ways} &= {}^{8}\mathrm{C}_{7} \times {}^{16}\mathrm{C}_{5} + {}^{8}\mathrm{C}_{8} \times {}^{16}\mathrm{C}_{4} \\ &= 8 \times 4368 + 1 \times 1820 \\ &= 36\,764.\end{aligned}$$

**PP 9** **Normanhurst Boys High School, 2023, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(e)(i)(ii)**

(i) Choosing 6 of the 12 people for one group, in ${}^{12}\mathrm{C}_{6} = 924$ ways, leaves the other 6 as the second group. But the two groups have no names, so each division has been counted twice: once for each group being the one chosen first. Hence

$$\text{number of ways} = \frac{{}^{12}\mathrm{C}_{6}}{2} = \frac{924}{2} = 462.$$

(ii) Put the two particular people in a group, and choose its other 4 members from the other 10 people. The rest form the second group. The group with the two particular people is now different from the other group, so nothing is counted twice:

$$\text{number of ways} = {}^{10}\mathrm{C}_{4} = 210.$$

**PP 10** **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(d)**

The number of groups of three is ${}^{n}\mathrm{C}_{3}$ and the number of groups of two is ${}^{n}\mathrm{C}_{2}$, where $n \geq 3$. We are told that

$$\begin{aligned}{}^{n}\mathrm{C}_{3} &= 3 \times {}^{n}\mathrm{C}_{2} \\ \frac{n(n-1)(n-2)}{6} &= 3 \times \frac{n(n-1)}{2}.\end{aligned}$$

Since $n \geq 3$, $n(n-1) \neq 0$, so we may divide both sides by $n(n-1)$:

$$\frac{n-2}{6} = \frac{3}{2}, \qquad n-2 = 9, \qquad n = 11.$$

There are 11 dancers. Check: ${}^{11}\mathrm{C}_{3} = 165 = 3 \times 55 = 3 \times {}^{11}\mathrm{C}_{2}$.

**PP 11** **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q12(c)**

Choose the 4 shaded tiles of the top row: ${}^{10}\mathrm{C}_4 = 210$ ways.

Choose which 2 of these 4 are above shaded tiles: ${}^{4}\mathrm{C}_2 = 6$ ways. The bottom tiles under these 2 are shaded. The bottom tiles under the other 2 top shaded tiles must not be shaded.

That leaves 4 more bottom tiles to shade, from the 6 bottom tiles that are not under a shaded top tile: ${}^{6}\mathrm{C}_4 = 15$ ways. Hence

$$\text{number of ways} = {}^{10}\mathrm{C}_4 \times {}^{4}\mathrm{C}_2 \times {}^{6}\mathrm{C}_4 = 210 \times 6 \times 15 = 18\,900.$$

# Chapter 14 Combinatorics - Homework Answers

Lesson 14F | Using counting in probability

## Lesson Examples

**C1**

**Tutor demo.**

**a** Unordered hands: ${}^{52}\mathrm{C}_4 = 270\,725$. Favourable: ${}^{13}\mathrm{C}_1 \times {}^{13}\mathrm{C}_3 = 13 \times 286 = 3718$. So $P = \dfrac{3718}{270\,725} = \dfrac{286}{20\,825}$.

**b** Ordered hands: ${}^{52}\mathrm{P}_4 = 52 \times 51 \times 50 \times 49$. Favourable: $13 \times 13 \times 12 \times 11$. So $P(\heartsuit\spadesuit\spadesuit\spadesuit) = \dfrac{13 \times 13 \times 12 \times 11}{52 \times 51 \times 50 \times 49} = \dfrac{143}{41\,650}$, which is $\frac{1}{4}$ of part **a**, because the heart can be in any one of four positions.

**Your turn.**

**a** $P = \dfrac{26 \times 26}{{}^{52}\mathrm{C}_2} = \dfrac{676}{1326} = \dfrac{26}{51}$

**b** $P = \dfrac{26 \times 26}{52 \times 51} = \dfrac{13}{51}$, which is $\frac{1}{2}$ of part **a**.

**C2**

**Tutor demo.** There are $9 \times 10 \times 10 \times 10 = 9000$ four-digit numbers.

**a** $\dfrac{3 \times 10 \times 10 \times 10}{9000} = \dfrac{1}{3}$

**b** $\dfrac{5 \times 5 \times 5 \times 5}{9000} = \dfrac{5}{72}$

**c** $\dfrac{9 \times 9 \times 8 \times 7}{9000} = \dfrac{63}{125}$

**d** $\dfrac{{}^{9}\mathrm{C}_4}{9000} = \dfrac{126}{9000} = \dfrac{7}{500}$

**Your turn.** There are $9 \times 10 \times 10 = 900$ three-digit numbers.

**a** $\dfrac{6 \times 10 \times 10}{900} = \dfrac{2}{3}$

**b** $\dfrac{4 \times 5 \times 5}{900} = \dfrac{1}{9}$

**c** $\dfrac{9 \times 9 \times 8}{900} = \dfrac{18}{25}$

**d** $\dfrac{{}^{9}\mathrm{C}_3}{900} = \dfrac{84}{900} = \dfrac{7}{75}$

**C3**

**Tutor demo.**

**a** Without a 3: $8 \times 9 \times 9 \times 9 = 5832$, so with a 3: $9000 - 5832 = 3168$, and $P = \dfrac{3168}{9000} = \dfrac{44}{125}$.

**b** No 3 and no 6: $7 \times 8 \times 8 \times 8 = 3584$. No 3 or no 6: $5832 + 5832 - 3584 = 8080$. So at least one 3 and at least one 6: $9000 - 8080 = 920$, and $P = \dfrac{920}{9000} = \dfrac{23}{225}$.

**c** Strings with exactly two 7s: ${}^4\mathrm{C}_2 \times 9 \times 9 = 486$. Those beginning with zero: ${}^3\mathrm{C}_2 \times 9 = 27$. So $486 - 27 = 459$ numbers, and $P = \dfrac{459}{9000} = \dfrac{51}{1000}$.

**d** Exactly three 7s: $*777$ gives 8, and $7*77$, $77*7$, $777*$ give 9 each, so $8 + 27 = 35$. Four 7s: 1. So $459 + 35 + 1 = 495$ numbers, and $P = \dfrac{495}{9000} = \dfrac{11}{200}$.

**Your turn.**

**a** Without a 5: $8 \times 9 \times 9 = 648$, so with a 5: $900 - 648 = 252$, and $P = \dfrac{252}{900} = \dfrac{7}{25}$.

**b** No 5 and no 6: $7 \times 8 \times 8 = 448$. No 5 or no 6: $648 + 648 - 448 = 848$. So at least one 5 and at least one 6: $900 - 848 = 52$, and $P = \dfrac{52}{900} = \dfrac{13}{225}$.

**c** Strings with exactly two 2s: ${}^3\mathrm{C}_2 \times 9 = 27$. Those beginning with zero: only 022, so 1. So $27 - 1 = 26$ numbers, and $P = \dfrac{26}{900} = \dfrac{13}{450}$.

**d** Add the one number 222 with three 2s: $26 + 1 = 27$ numbers, and $P = \dfrac{27}{900} = \dfrac{3}{100}$.

## Textbook Homework

**1** **a** $84$ **b** $\dfrac{5}{42}$

**2** **a** $\dfrac{1}{210}$ **b** $\dfrac{2}{5}$ **c** $\dfrac{3}{5}$ **d** $\dfrac{4}{15}$

**3** **a** $\dfrac{1}{13}$ **b** $\dfrac{46}{455}$ **c** $\dfrac{3}{91}$ **d** $\dfrac{3}{13}$

**4** **a** $\dfrac{8}{429}$ **b** $\dfrac{1}{143}$ **c** $\dfrac{140}{429}$ **d** $\dfrac{421}{429}$ **e** $\dfrac{2}{11}$ **f** $\dfrac{1}{3}$

5 a $\frac{1}{22\,100}$ b $\frac{1}{5525}$ c $\frac{11}{850}$ d $\frac{22}{425}$ e $\frac{11}{1105}$
f $\frac{13}{34}$ g $\frac{16}{5525}$ h $\frac{6}{5525}$ i $\frac{741}{1700}$ j $\frac{64}{425}$

6 a $\frac{3}{70\,304}$ b $\frac{1}{2197}$ c $\frac{1}{64}$ d $\frac{1}{16}$ e $\frac{27}{2197}$ f $\frac{3}{8}$ g $\frac{6}{2197}$ h $\frac{3}{2197}$ i $\frac{27}{64}$ j $\frac{5}{32}$

7 a $\frac{1}{10}$ b $\frac{1}{10}$ c $\frac{1}{3}$

8 a $\frac{1}{10}$ b $\frac{2}{5}$

9 a $\frac{1}{15}$ b $\frac{2}{3}$

10 a $\frac{1}{42}$ b $\frac{2}{7}$ c $\frac{2}{7}$ d $\frac{1}{35}$ e $\frac{1}{7}$

11 a $\frac{1}{2}$ b $\frac{1}{6}$ c $\frac{1}{5}$ d $\frac{1}{60}$ e $\frac{2}{3}$

12 a $\frac{1}{7}$ b $\frac{2}{7}$ c $\frac{1}{7}$ d $\frac{2}{7}$

13 a $\frac{1}{3}$ b $\frac{2}{3}$ c $\frac{2}{15}$ d $\frac{1}{5}$

14 a $\frac{1}{26}$ b $\frac{5}{13}$ c $\frac{15}{26}$ d $\frac{1}{26}$

15 a $0.403$ b $0.597$ c $0.001\,74$ d $0.291$

16 a $\frac{1}{22}$ b $\frac{125}{1728}$ c $\frac{5}{144}$

17 a $\frac{10}{31}$ b $\frac{15}{31}$ c $\frac{6}{31}$

18 a $\frac{1}{60}$ b $\frac{3}{5}$ c $\frac{1}{5}$ d $\frac{2}{5}$ e $\frac{1}{20}$ f $\frac{3}{5}$ g $\frac{3}{10}$ h $\frac{9}{10}$ i $\frac{1}{10}$ j $\frac{3}{5}$

19 a $\frac{3}{8}$ b $\frac{1}{2}$ c $\frac{21}{32}$ d $\frac{3}{32}$ e $\frac{17}{64}$

20 a $\frac{281}{462}$ b $N = 8$

21 a $\frac{1}{27\,417}$ b $\frac{28}{703}$

**22** In each part, the sample space has ${}^{52}\mathrm{C}_5$ members.

**a** $\dfrac{352}{833}$. Choose the value of the pair in 13 ways, then choose the cards in the pair in ${}^4\mathrm{C}_2 = 6$ ways, then choose the three values of the three remaining cards in ${}^{12}\mathrm{C}_3$ ways, then choose the suits of those three cards in $4^3$ ways.

**b** $\dfrac{198}{4165}$. Choose the values of the two pairs in ${}^{13}\mathrm{C}_2$ ways, then choose the suits of the cards in the two pairs in ${}^4\mathrm{C}_2 \times {}^4\mathrm{C}_2$ ways, then choose the remaining card in 44 ways.

**c** $\dfrac{88}{4165}$ **d** $\dfrac{1}{4165}$ **e** $\dfrac{6}{4165}$

**f** $\dfrac{128}{32\,487}$. Choose the lowest card in 10 ways, then choose the suits of the five cards in $4^5$ ways.

**g** $\dfrac{33}{16\,660}$ **h** $\dfrac{1}{649\,740}$

**23** **a** $\dfrac{1}{125}$ **b** $\dfrac{4}{125}$ **c** $\dfrac{16}{125}$ **d** $\dfrac{108}{125}$

**24** **a** $\dfrac{48}{125}$ **b** $\dfrac{n^2(n-1)^2(n-2)!}{2n^n}$

**25** **a** $0.0082$ **b** $1 - \dfrac{{}^{365}\mathrm{P}_n}{365^n}$ **d** $23$ **e** $41$

**26** **a** $\dfrac{1}{25}$ **b** $\dfrac{3}{25}$ **c** $\dfrac{19}{25}$

**27** **b** $\dfrac{1}{8}$ **c** $2^{1-n}$

# Chapter 14 Combinatorics - Homework Answers

Lesson 14F | Using counting in probability

## Past-Paper Homework - Worked Solutions

**PP 1** **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q8**

The marbles of one colour look alike, but they are still 14 different marbles, so every unordered selection of 3 of them is equally likely. There are $^{14}\mathrm{C}_3 = 364$ such selections.

For exactly two blue marbles, choose 2 of the 5 blue marbles in $^{5}\mathrm{C}_2 = 10$ ways, and the third marble from the $6 + 3 = 9$ marbles that are not blue in $^{9}\mathrm{C}_1 = 9$ ways. Hence

$$P(\text{exactly two blue}) = \frac{^{5}\mathrm{C}_2 \times {}^{9}\mathrm{C}_1}{^{14}\mathrm{C}_3} = \frac{90}{364} = \frac{45}{182}.$$

**Answer (B).** Option (A) forgets to choose the third marble, which must not be blue.

**PP 2** **Killara High School, 2023, Year 12 Examination, Mathematics Extension 1 Trial, Q6**

The draw is an unordered selection of 6 of the 30 numbers, so there are $^{30}\mathrm{C}_6 = 593\,775$ equally likely draws.

A draw matches exactly $k$ of the player's numbers when it contains $k$ of the player's 6 numbers and $6 - k$ of the other 24 numbers. 'At least 4' means 4, 5 or 6 matches, so

$$\begin{aligned}\text{number of favourable draws} &= {}^{6}\mathrm{C}_4 \times {}^{24}\mathrm{C}_2 + {}^{6}\mathrm{C}_5 \times {}^{24}\mathrm{C}_1 + {}^{6}\mathrm{C}_6 \times {}^{24}\mathrm{C}_0 \\ &= 15 \times 276 + 6 \times 24 + 1 \\ &= 4285.\end{aligned}$$

$$\text{Hence} \qquad P(\text{at least 4 matches}) = \frac{4285}{593\,775} \approx 0.0072.$$

**Answer B.** To three decimal places the probability is 0.007.

**PP 3** **Normanhurst Boys High School, 2020, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(f)**

A team is an unordered selection. The sample space is the set of all teams of 13 chosen from the 25 players, so it has $^{25}\mathrm{C}_{13} = 5\,200\,300$ members. A team of only female players is a choice of 13 of the 20 female players, in $^{20}\mathrm{C}_{13} = 77\,520$ ways. Hence

$$P(\text{only female players}) = \frac{^{20}\mathrm{C}_{13}}{^{25}\mathrm{C}_{13}} = \frac{77\,520}{5\,200\,300} = \frac{12}{805}.$$

**PP 4** **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q11(c)**

The two cards form an unordered hand, so the sample space has ${}^{52}\mathrm{C}_2 = 1326$ members. 'At least 1 heart' is a warning, so consider the complement, 'no heart'. Then both cards come from the 39 cards that are not hearts, in ${}^{39}\mathrm{C}_2 = 741$ ways. Hence

$$P(\text{at least 1 heart}) = 1 - \frac{{}^{39}\mathrm{C}_2}{{}^{52}\mathrm{C}_2} = 1 - \frac{741}{1326} = \frac{15}{34}.$$

Check with ordered selections: $1 - \dfrac{39 \times 38}{52 \times 51} = \dfrac{15}{34}$.

**PP 5** **Hornsby Girls High School, 2022, Mathematics Extension 1, Year 12 Higher School Certificate Trial Examination Term 3, Q12(c)**

There are $10 + 8 = 18$ students, so there are ${}^{18}\mathrm{C}_5 = 8568$ possible committees of 5. Choose the 3 girls in ${}^{10}\mathrm{C}_3 = 120$ ways and, separately, the 2 boys in ${}^{8}\mathrm{C}_2 = 28$ ways. Hence

$$P(3 \text{ girls and } 2 \text{ boys}) = \frac{{}^{10}\mathrm{C}_3 \times {}^{8}\mathrm{C}_2}{{}^{18}\mathrm{C}_5} = \frac{120 \times 28}{8568} = \frac{20}{51}.$$

**PP 6** **Normanhurst Boys High School, 2022, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(c)(i)(ii)**

(i) A committee is an unordered selection of 7 of the 20 prefects:

$$\text{number of committees} = {}^{20}\mathrm{C}_7 = 77\,520$$

(ii) Jun is on the committee and Alex is not, so the other 6 members are chosen from the 18 prefects other than Jun and Alex, in ${}^{18}\mathrm{C}_6 = 18\,564$ ways. Hence

$$P(\text{Jun but not Alex}) = \frac{{}^{18}\mathrm{C}_6}{{}^{20}\mathrm{C}_7} = \frac{18\,564}{77\,520} = \frac{91}{380}.$$

**PP 7** **Girraween High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q11(c)(i)(ii)**

The division is fixed once the 20 students in class A are chosen, because class B is the other 20. So there are ${}^{40}\mathrm{C}_{20}$ equally likely divisions.

(i) If all four friends are in class A, the other 16 students in A are chosen from the other 36 students, in ${}^{36}\mathrm{C}_{16}$ ways. If all four are in class B, class A is 20 of the other 36 students, in ${}^{36}\mathrm{C}_{20} = {}^{36}\mathrm{C}_{16}$ ways. Since

$$\frac{{}^{36}\mathrm{C}_{16}}{{}^{40}\mathrm{C}_{20}} = \frac{36!}{16! \times 20!} \times \frac{20! \times 20!}{40!} = \frac{20 \times 19 \times 18 \times 17}{40 \times 39 \times 38 \times 37} = \frac{51}{962},$$

$$P(\text{all four in the same class}) = \frac{2 \times {}^{36}\mathrm{C}_{16}}{{}^{40}\mathrm{C}_{20}} = \frac{51}{481} \approx 0.106.$$

(ii) For exactly three friends in class A, choose the three in ${}^{4}\mathrm{C}_{3} = 4$ ways. The fourth friend is in class B, so the other 17 students in A are chosen from the other 36 students, in ${}^{36}\mathrm{C}_{17}$ ways. Exactly three in class B is counted in the same way. Since

$$\frac{{}^{36}\mathrm{C}_{17}}{{}^{40}\mathrm{C}_{20}} = \frac{36!}{17! \times 19!} \times \frac{20! \times 20!}{40!} = \frac{20 \times 19 \times 18 \times 20}{40 \times 39 \times 38 \times 37} = \frac{30}{481},$$

$$P(\text{exactly three in the same class}) = \frac{2 \times {}^{4}\mathrm{C}_{3} \times {}^{36}\mathrm{C}_{17}}{{}^{40}\mathrm{C}_{20}} = \frac{240}{481} \approx 0.499.$$

# Chapter 14 Combinatorics - Homework Answers

Chapter 14 Review | Chapter review exercise

1 $8! = 40\,320$

2 **a** $72$ **b** $n(n+1)$ **c** $k \times k!$

3 **a** $792$ **b** $6$ **c** $1140$

4 ${}^{7}\mathrm{P}_{4} = 840$

5 $26^3 \times 10^4 = 175\,760\,000$

6 **a** $4! \times 3! = 144$ **b** $5! = 120$ **c** $3! \times 4! = 144$ **d** $7! \div 2 = 2520$ (half of the total)

7 **a** $10! = 3\,628\,800$ **b** $2 \times 5! \times 5! = 28\,800$ **c** $9! \times 2! = 725\,760$

8 $\dfrac{8!}{3! \times 2!} = 3360$

9 ${}^{6}\mathrm{P}_{3} + {}^{6}\mathrm{P}_{4} = 480$

10 $\dfrac{10!}{6! \times 4!} = 210$

**11** **a** $^{16}\mathrm{C}_7 = 11\,440$ **b** $^{10}\mathrm{C}_7 = 120$ **c** $0$ (there are no such committees of 7)
**d** $^{6}\mathrm{C}_2 \times {}^{10}\mathrm{C}_5 = 3780$ **e** $^{10}\mathrm{C}_4 \times {}^{6}\mathrm{C}_3 = 4200$
**f** By adding cases: $^{10}\mathrm{C}_4 \times {}^{6}\mathrm{C}_3 + {}^{10}\mathrm{C}_5 \times {}^{6}\mathrm{C}_2 + {}^{10}\mathrm{C}_6 \times {}^{6}\mathrm{C}_1 + {}^{10}\mathrm{C}_7 = 9360$
**g** $^{15}\mathrm{C}_6 = 5005$ **h** The complement of part **g**, that is, $^{15}\mathrm{C}_7 = 6435$
**i** The complement of the case where both are members, that is, $^{16}\mathrm{C}_7 - {}^{14}\mathrm{C}_5 = 9438$

**12** **a** $^{8}\mathrm{C}_5 = 56$ **b** $^{8}\mathrm{C}_4 = 70$ **c** $35$

**13** More men than women means four men and two women. Choose the four men members, then choose the two women members: $\dfrac{^{4}\mathrm{C}_4 \times {}^{3}\mathrm{C}_2}{^{7}\mathrm{C}_6} = \dfrac{3}{7}$. Alternatively, only one of the seven is not on the committee, and there will be a majority of men when this person is a woman. Thus the probability is $\dfrac{3}{7}$.

**14** **a** $\dfrac{1}{22\,100}$ **b** $\dfrac{1}{5525}$ **c** $\dfrac{11}{850}$ **d** $\dfrac{22}{425}$ **e** $\dfrac{13}{34}$ **f** $\dfrac{16}{5525}$ **g** $\dfrac{6}{5525}$ **h** $\dfrac{64}{425}$